

Heterogeneity Discussion

What is the goal?

- Conveying heterogeneity to consumers of structures (mmCIF files)

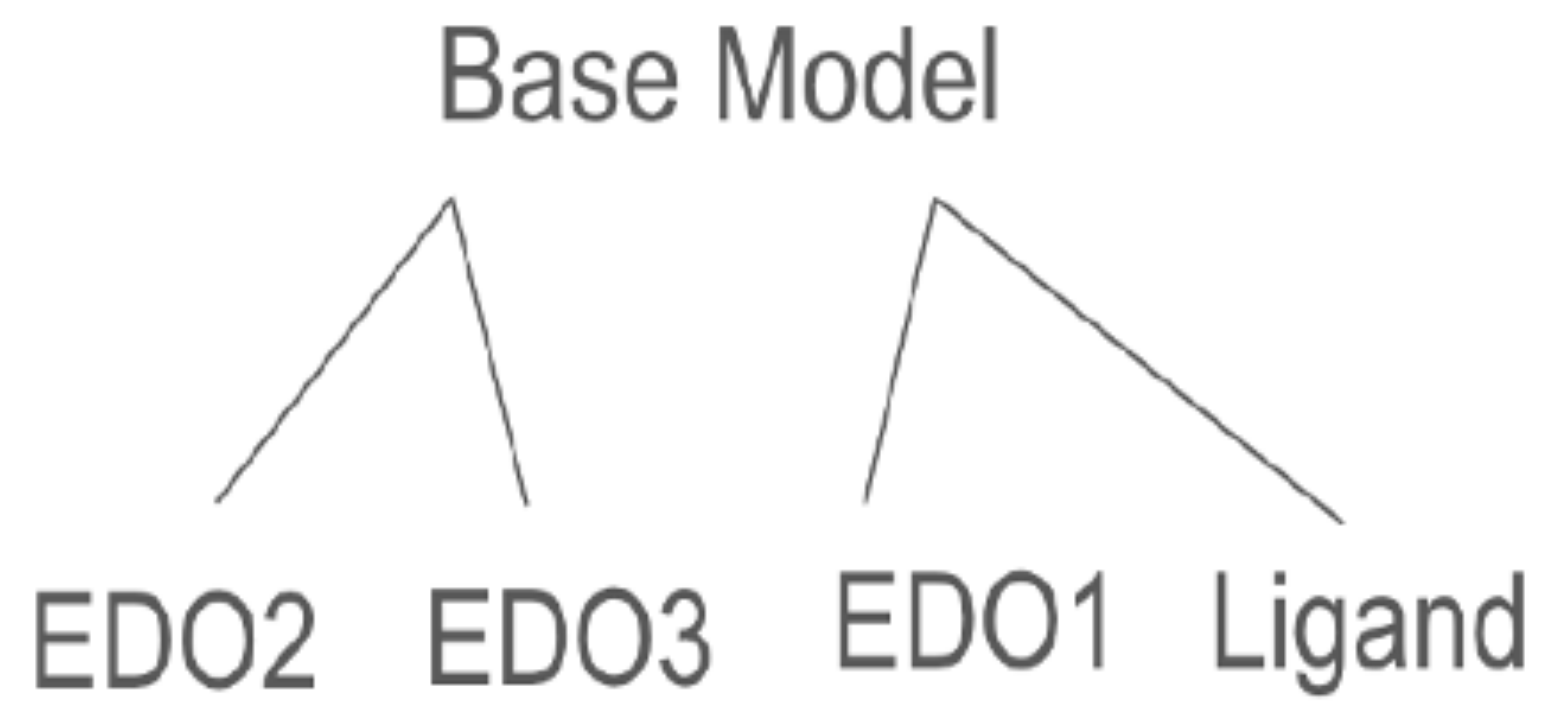
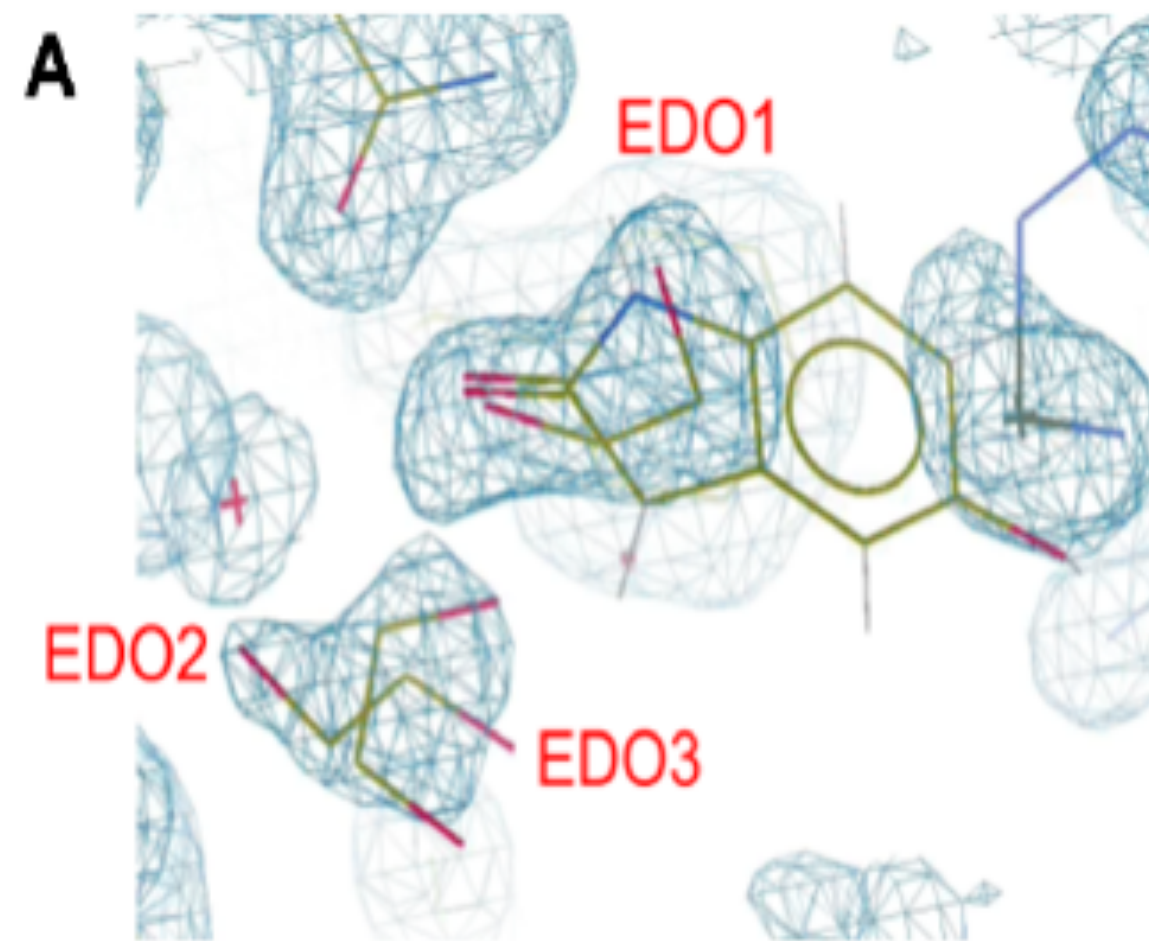
and

- Performing calculations on models with heterogeneity?

What is the goal?

- Conveying heterogeneity to consumers of structures (mmCIF files) can be achieved by adding information to the mmCIF files - but would require some thought as to how structures would be visualized
- Performing calculations on models with heterogeneity requires interpretation of the heterogeneity definition - which needs to be unambiguous, with a common understanding across software systems

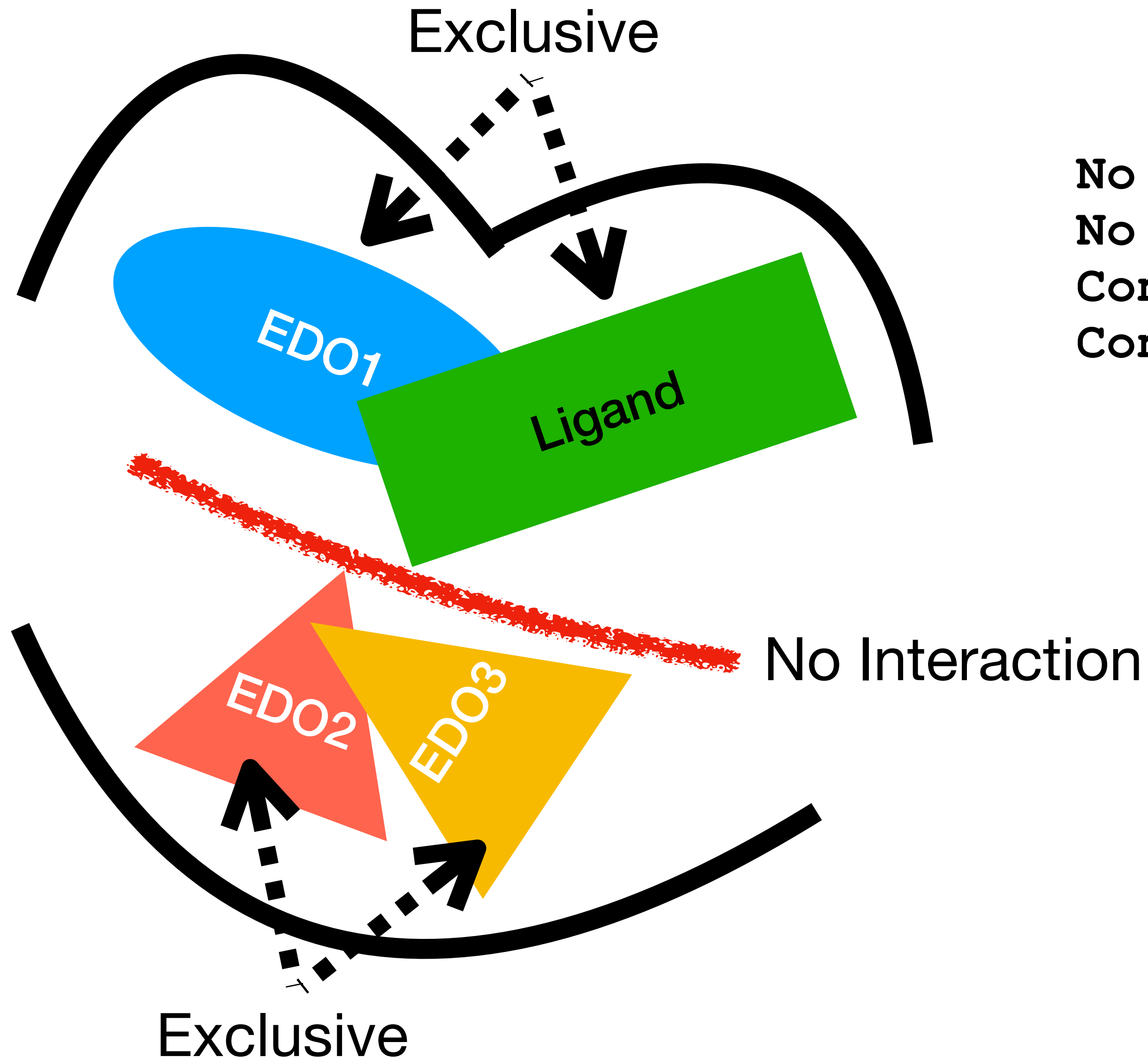
An Example



```
loop_  
_pdbx_heterogeneity_hierarchy.{name,parent,id,details}  
Base_name . Base ?  
EDO1_bound Base B ?  
Fragment_bound Base C ?  
EDO2_bound Base D ?  
EDO3_bound Base E ?
```

```
loop_  
_pdbx_state_coexistence.{id,rule,hets_id,hets_ids,description}  
1 NOT B C EDO1_bound_NOT_Fragment  
2 OR C D,E Fragment_OR_(EDO2,EDO3)  
3 OR B D,E EDO1_OR_(EDO2,EDO3)  
4 NOT D E EDO2_bound_NOT_EDO3_bound
```

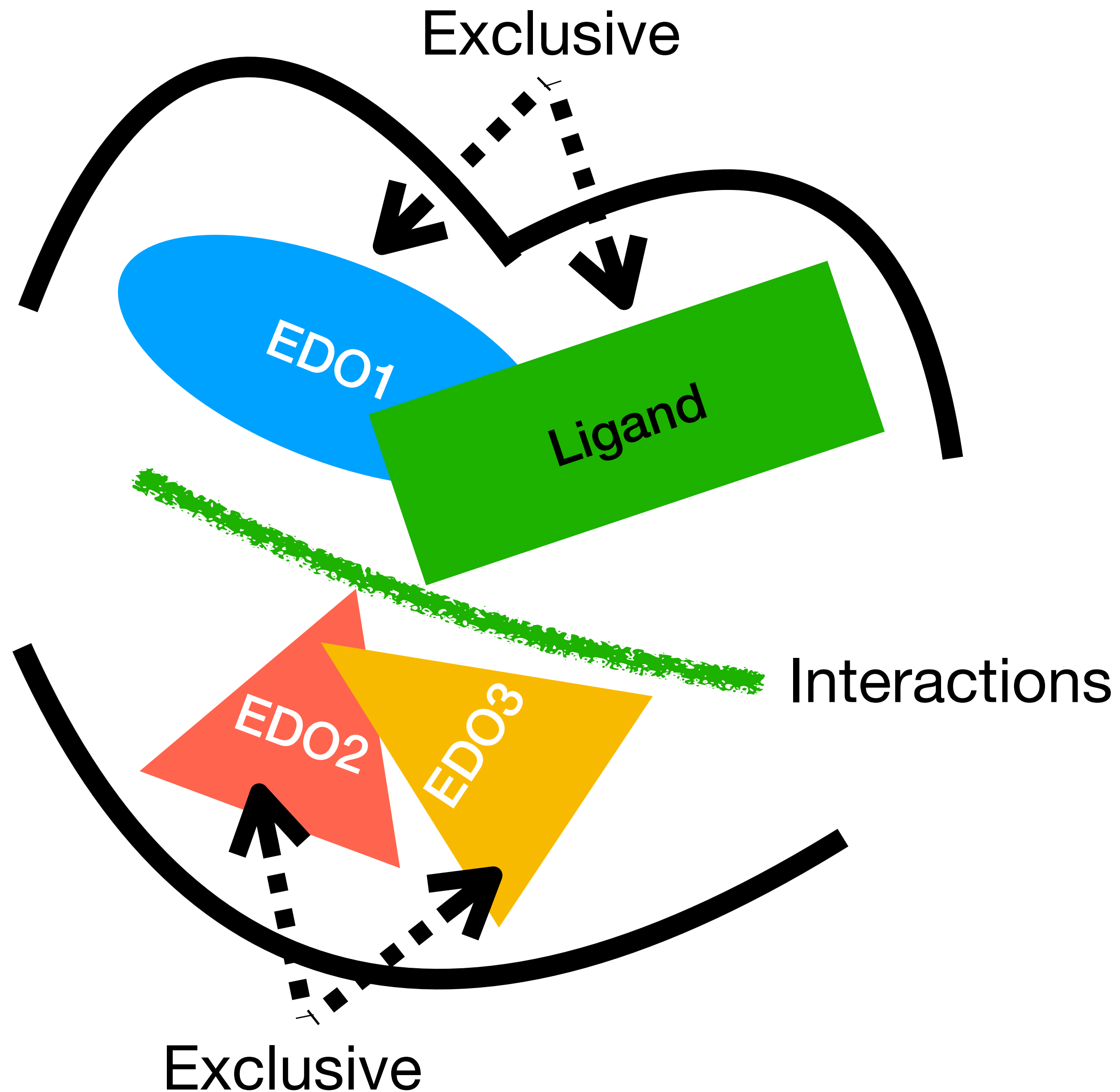
Interpretation 1



No interactions: Ligand and EDO1
No interactions: EDO2 and EDO3
Constrained occupancy: Ligand and EDO1
Constrained occupancy: EDO2 and EDO3

Ligand	(resid 1)	ALTLOC A
EDO1	(resid 1)	ALTLOC B
EDO2	(resid 2)	ALTLOC C
EDO3	(resid 2)	ALTLOC D

Interpretation 2



No interactions: Ligand and EDO1

No interactions: EDO2 and EDO3

Interactions: Ligand and EDO2

Interactions: Ligand and EDO3

Interactions: EDO1 and EDO2

Interactions: EDO1 and EDO3

Constrained occupancy: Ligand and EDO1

Constrained occupancy: EDO2 and EDO3

Constrained occupancy: Ligand and EDO2

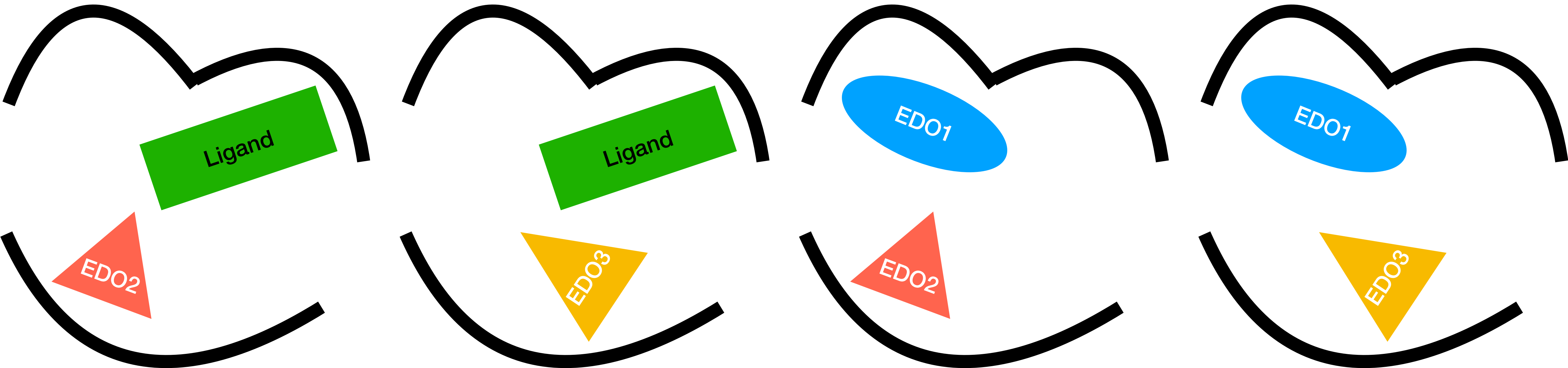
Constrained occupancy: Ligand and EDO3

Constrained occupancy: EDO1 and EDO2

Constrained occupancy: EDO1 and EDO3

Interpretation 2

Defining the different constituents across the crystal



Ligand (resid 1) ALTLOC A
EDO2 (resid 1) ALTLOC A

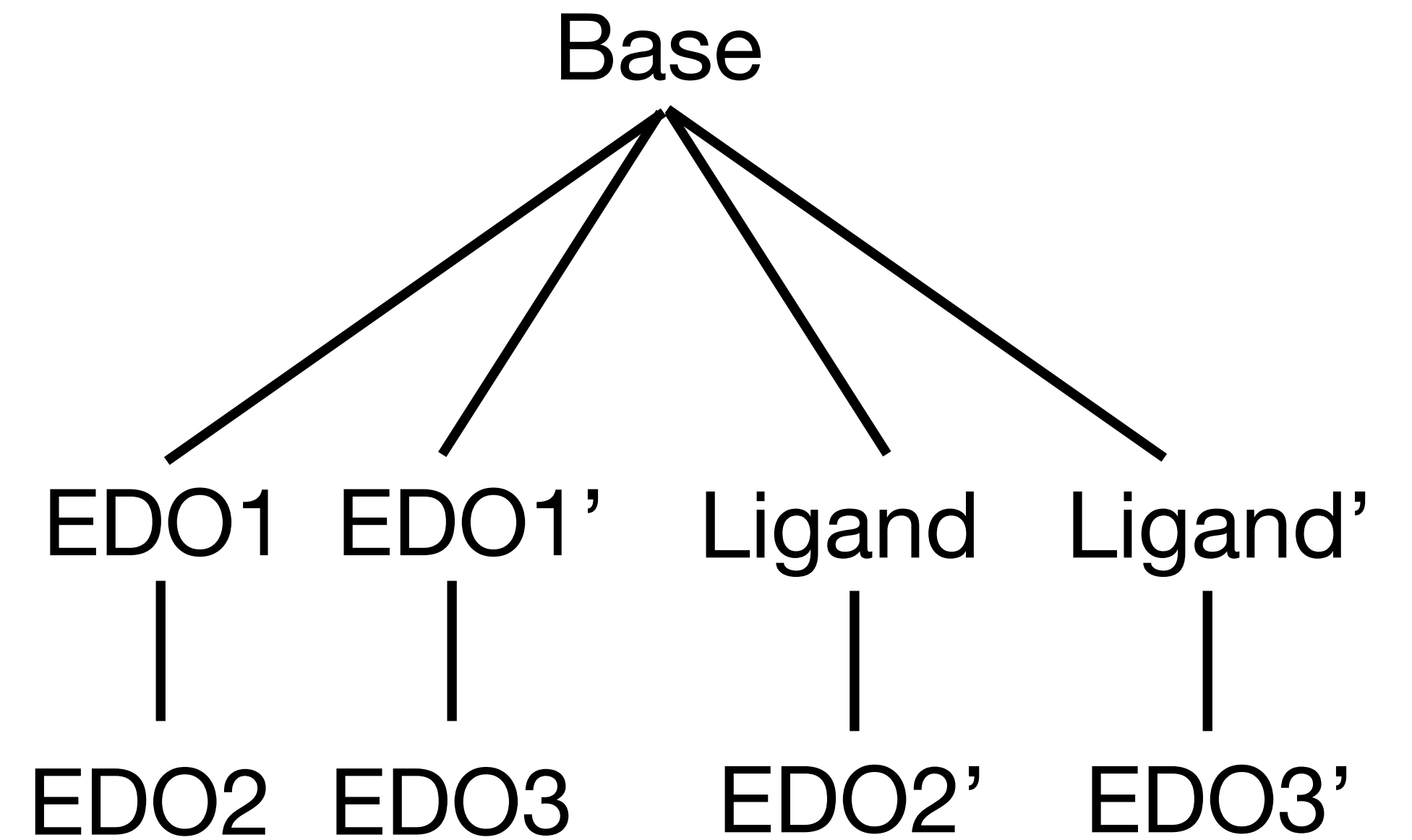
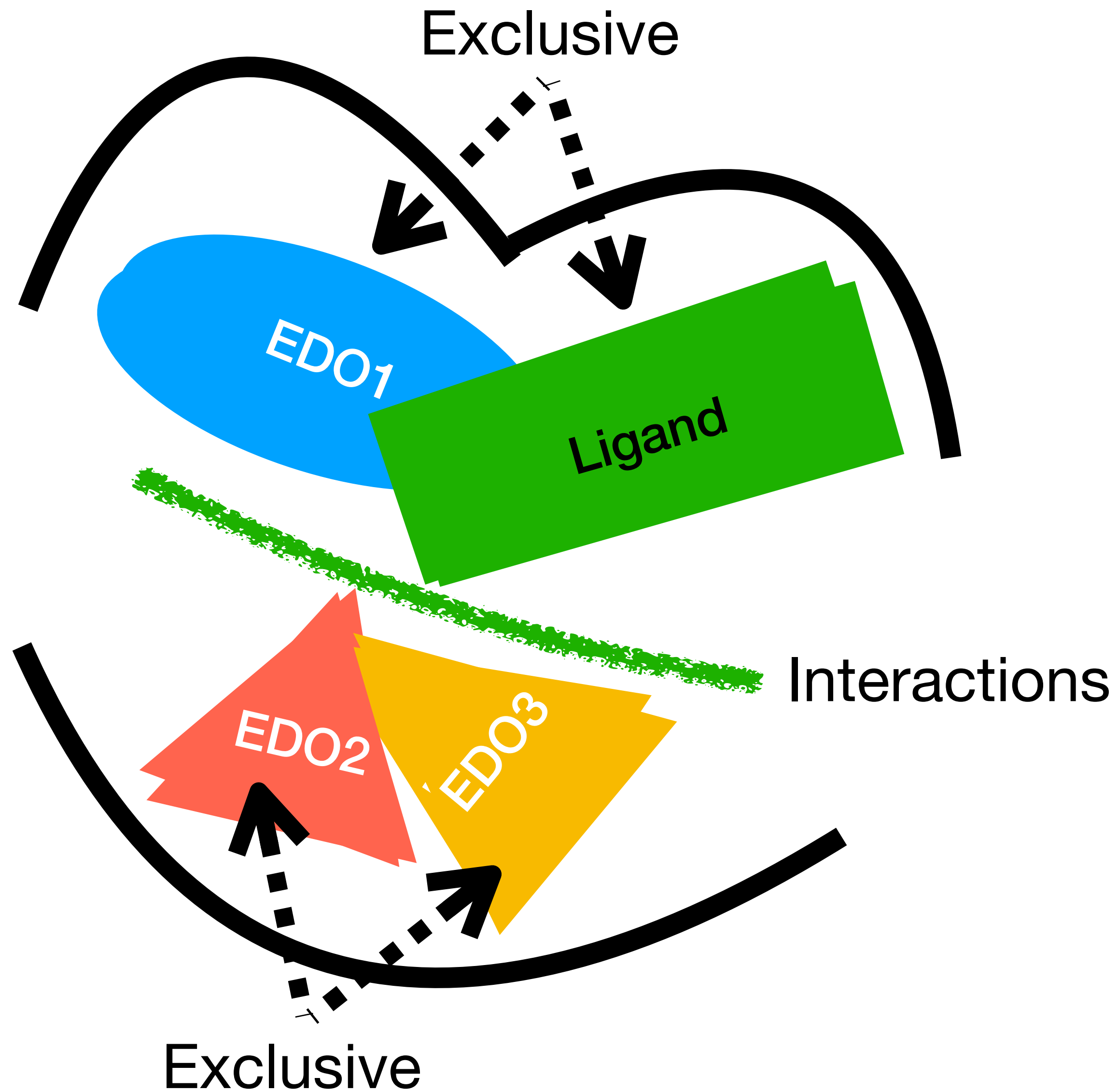
Ligand (resid 1) ALTLOC B
EDO3 (resid 1) ALTLOC B

EDO1 (resid 1) ALTLOC C
EDO2 (resid 1) ALTLOC C

EDO1 (resid 1) ALTLOC D
EDO3 (resid 1) ALTLOC D

The model has 2 copies of each small molecule

Interpretation 2



Interpretation 2

However, the presence of single species is undefined



Ligand (resid 1) ALTLOC E

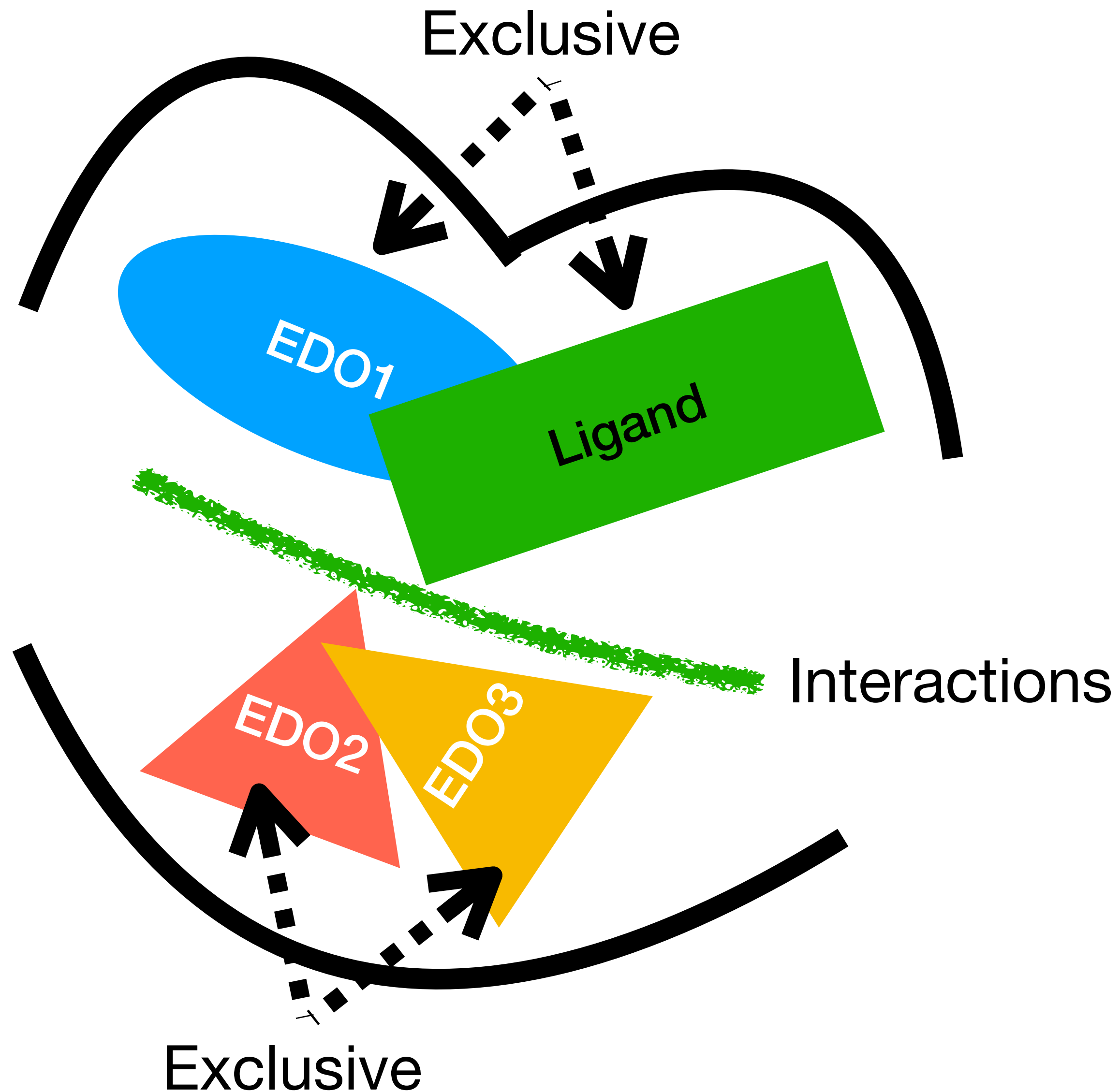
Ligand (resid 1) ALTLOC F

EDO1 (resid 1) ALTLOC G

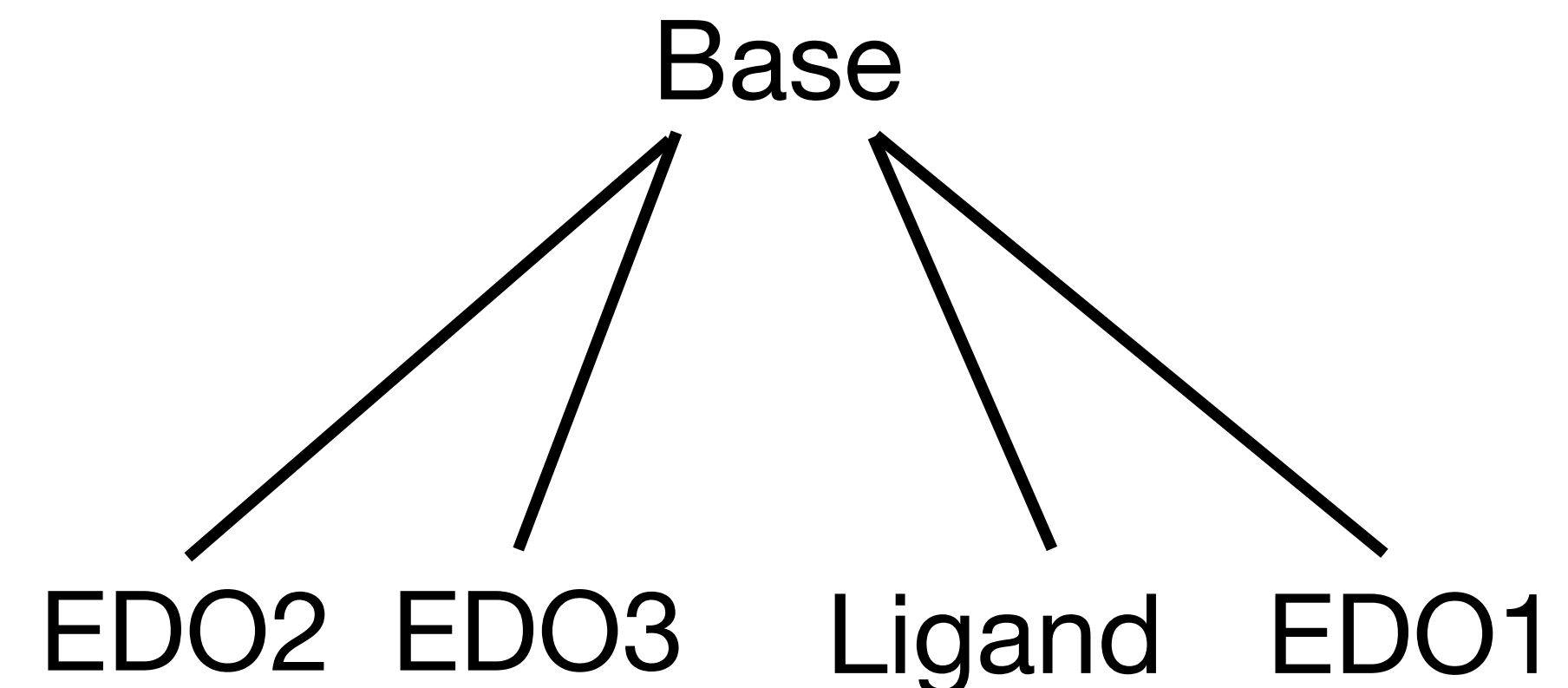
EDO1 (resid 1) ALTLOC H

The model has 3 copies of each small molecule

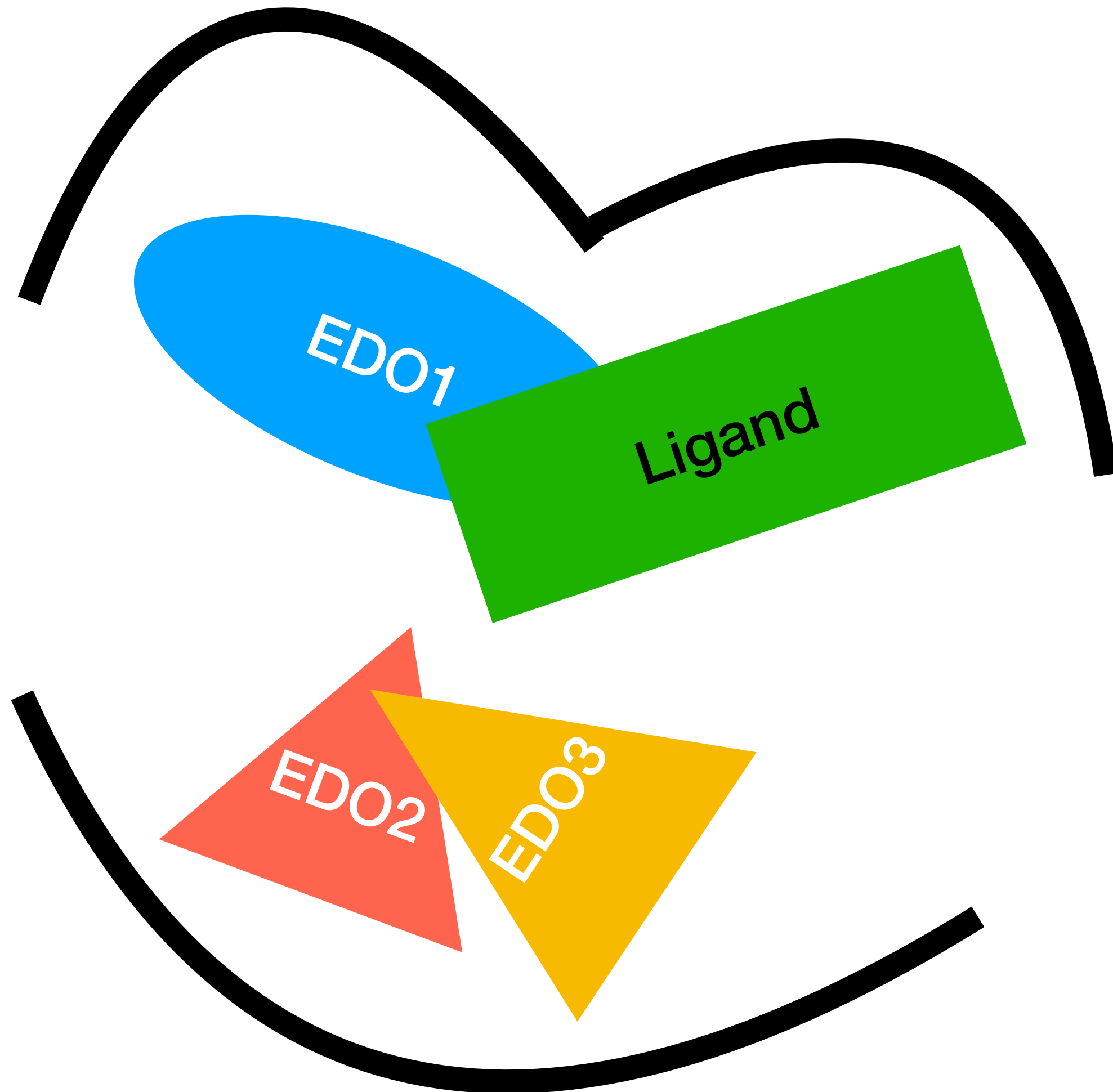
Interpretation 3



No interactions: Ligand and EDO1
No interactions: EDO2 and EDO3
Interactions: Ligand and EDO2
Interactions: Ligand and EDO3
Interactions: EDO1 and EDO2
Interactions: EDO1 and EDO3
Constrained occupancy: Ligand and EDO1
Constrained occupancy: EDO2 and EDO3
Constrained occupancy: Ligand and EDO2
Constrained occupancy: Ligand and EDO3
Constrained occupancy: EDO1 and EDO2
Constrained occupancy: EDO1 and EDO3



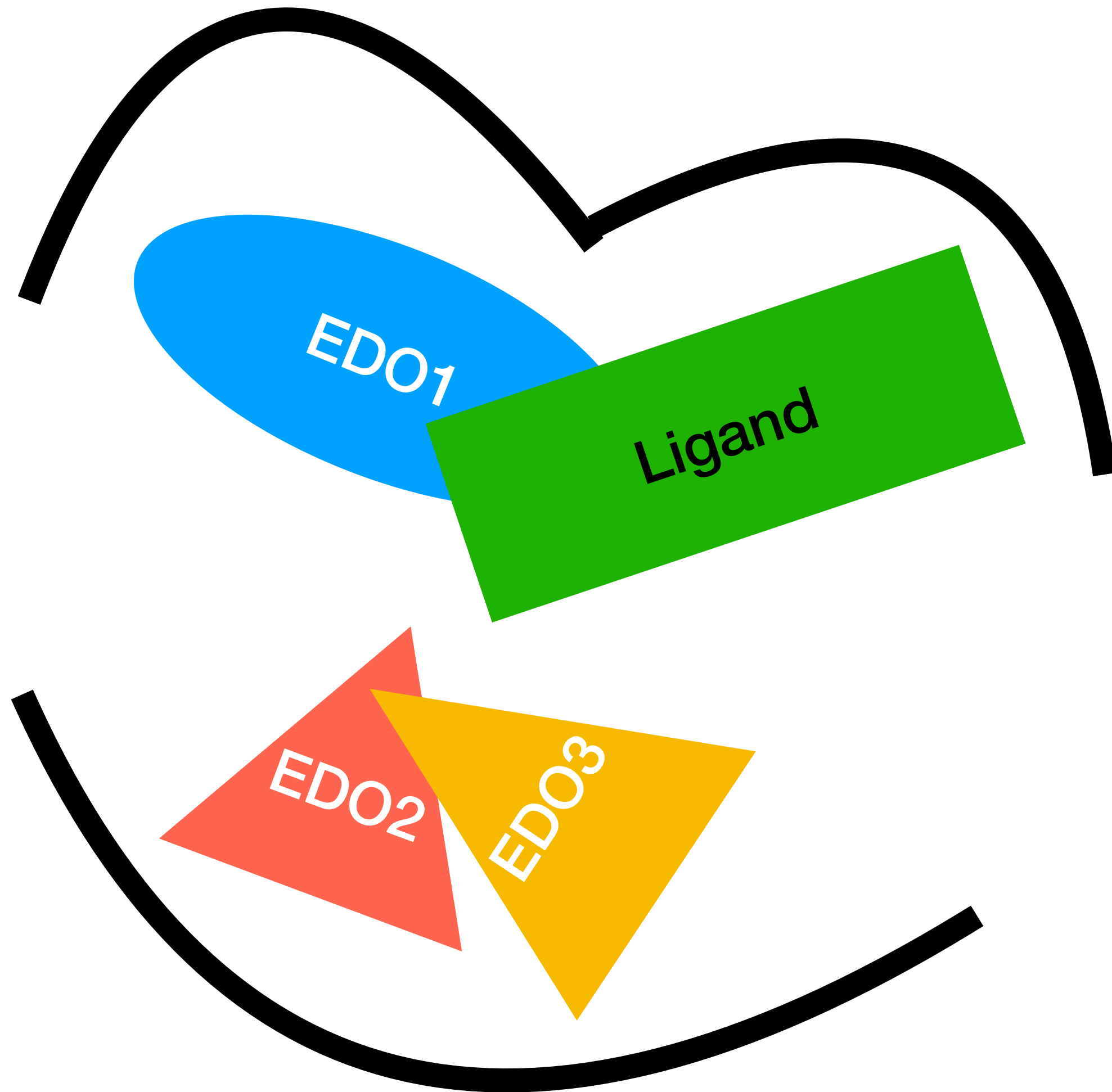
Interpretation 3



```
loop_  
_pdbx_heterogeneity_hierarchy.{name,parent,id,details}  
Base_name      .      Base ?  
EDO1_bound     Base  A    ?  
Fragment_bound Base  B    ?  
EDO2_bound     Base  C    ?  
EDO3_bound     Base  D    ?
```

```
loop_  
_pdbx_state_coexistence.{id,rule,hets_id,hets_ids,description}  
1 NOT  A  B    ?  
2 OR   A  C,D  ?  
3 OR   B  C,D  ?  
4 NOT  C  D    ?
```

Interpretation 3



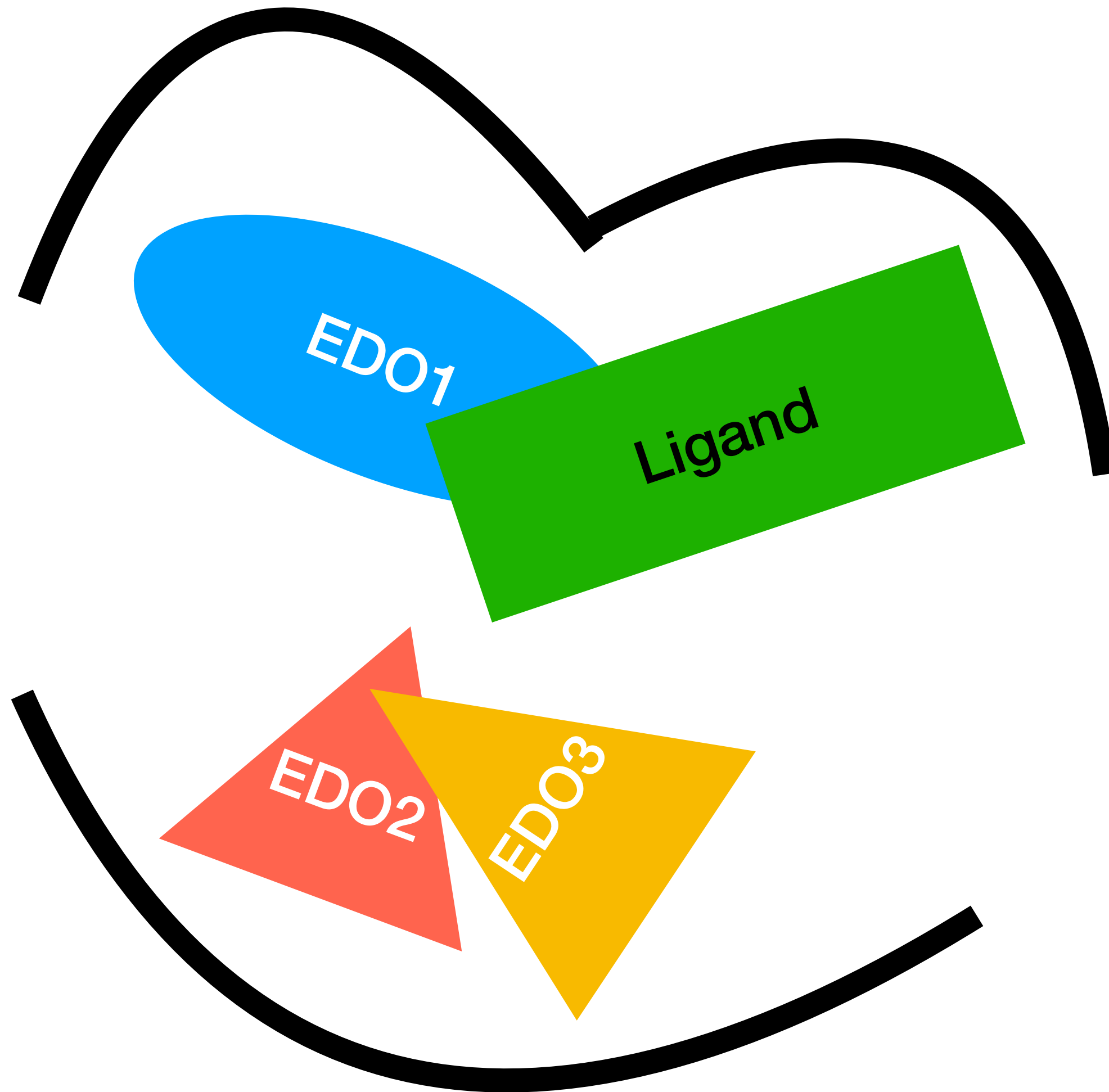
Assumptions:

- NOT implies occupation of a common site
- OR implies some kind of interaction

What can be deduced from the tables:

- EDO1 and Ligand see Base but not each other
- EDO2 and EDO3 see Base but not each other
- EDO1 interacts with EDO2
- EDO1 interacts with EDO3
- Ligand interacts with EDO2
- Ligand interacts with EDO3

Interpretation 3



Assumptions:

- NOT implies occupation of a common site
- OR implies some kind of interaction

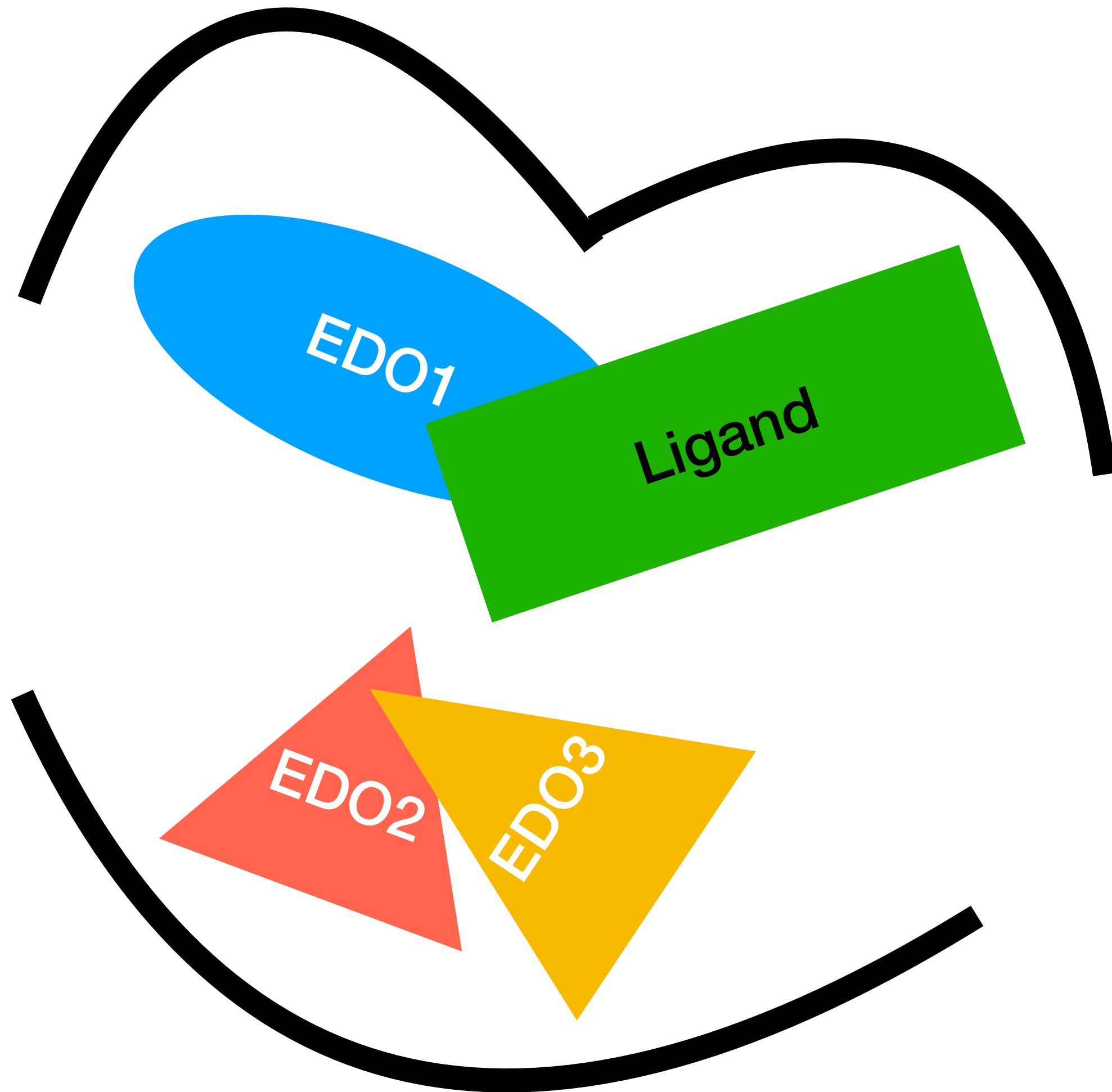
What can be deduced from the tables:

- Occupancy of EDO1 and Ligand are constrained
- Occupancy of EDO2 and EDO3 are constrained

What is unclear from the tables:

- Occupancy relationships between Ligand and EDO2 and EDO3
- Occupancy relationships between EDO1 and EDO2 and EDO3
- Existence of single species

Interpretation 3

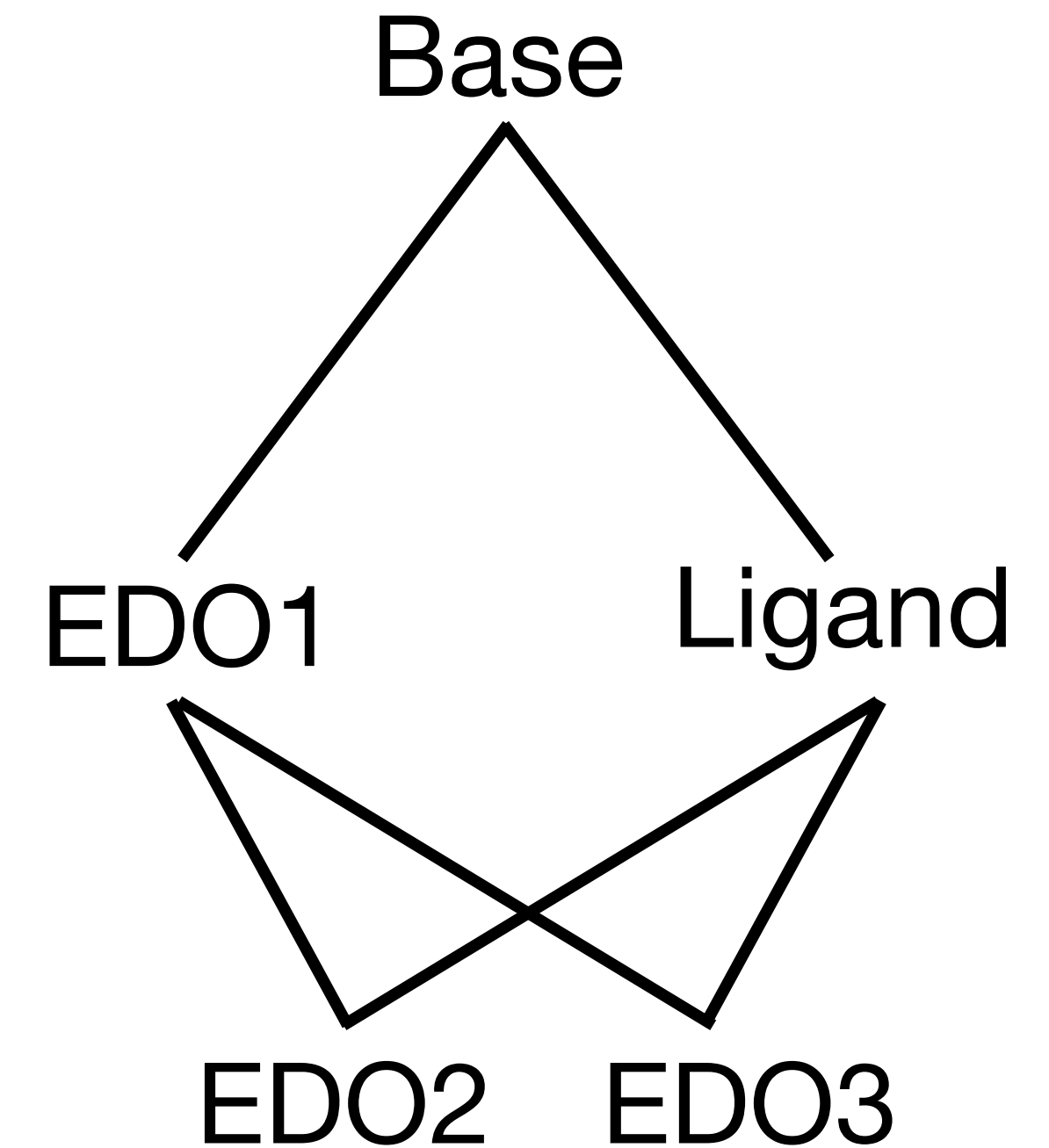


Possible Occupancy constraints:

- $O(\text{EDO1}) + O(\text{Ligand}) = 1$ (or $0 \leq 1$)
- $O(\text{EDO2}) + O(\text{EDO3}) = 1$ (or $0 \leq 1$)
- $O(\text{EDO1}) = O(\text{EDO2}) + O(\text{EDO3})$
- $O(\text{Ligand}) = O(\text{EDO2}) + O(\text{EDO3})$

What is Needed?

- The information about interactions could be retrieved from the hierarchy:
 - Branching indicates a common site
 - Connections indicate interactions
 - Interactions are inherited
- More information is needed to define occupancy relationships



This seems to be what is intended

Does that make sense?

Conclusions

- More information is required to make the intent unambiguous
 - The software systems need this to give the comparable results
- The intent dictates how interactions and occupancies are handled in refinement - with potentially major consequences for code